

MediAlert - Medication Reminder & Drug Interaction Checker

GROUP MEMBERS: MUHAMMAD HASNAT (241814), AZLAN ZAFAR
(241820), ANSHARA ZAHOOR (241802)
COURSE: VISUAL PROGRAMMING LAB
SUBMITTED TO: SIR OBAID ULLAH KHAN
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Reading Guide:

Each row in the requirement tables below follows the standard User Story format:

- There are two sections: one for functional requirements and one for non-functional requirements.
- ID — Unique requirement identifier (FR-xx for Functional, NFR-xx for Non-Functional)
- Role Badge — The user role the story belongs to: Patient, Caregiver, Doctor, Admin, or System
- User Story — Written in the format: As a [Role], I want [goal] so that [benefit]
- Acceptance Criteria (green) — Conditions that MUST be true for the story to be considered complete
- Rejection Criteria (red) — Conditions whose presence indicates the story has FAILED

Role color key: [Patient] [Caregiver] [Doctor] [Admin] [System]

Section 1:

Functional Requirements:

Functional requirements specify what the MediAlert system must do. They are written as user stories capturing the needs of all four roles: Patient, Caregiver, Doctor, and Admin.

1.1 Module 1: Authentication & User Management

Covers user registration, login, email verification, role assignment, profile management, and admin account control.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-01	[Patient] As a Patient, I want to register an account with my email and password so that I can access MediAlert's medication management features.	✓ Registration form accepts email, password, and role selection ✓ Passwords are stored as bcrypt hashes, never plaintext	✗ Registration proceeds without email storage ✗ Duplicate email addresses are accepted ✗ Weak passwords (< 8 chars) are accepted without warning
FR-02	[Patient] As a Patient, I want to log in with my email and password to access my personal medication dashboard securely .	✓ JWT token issued upon successful login ✓ Login fails gracefully with a clear message for wrong credentials ✓ Email-unverified accounts are blocked from login with an explanation	✗ Login succeeds with an unsaved email ✗ Plain-text credentials are transmitted without hashing. ✗ No session expiry is enforced
FR-03	[Admin] As an Admin, I want to suspend or delete any user account so that I can maintain platform safety and remove malicious or inactive users.	✓ Admin can search users by name or email ✓ Suspended users cannot log in and see a 'suspended' message ✓ Deletion requires a confirmation dialog before executing ✓ Deletion is blocked if active subscription/payment records exist	✗ Admins can be suspended or deleted by another Admin through the same UI ✗ User data is purged without a confirmation step
FR-04	[Patient] As a Patient, I want to update my personal profile data so that my	✓ All editable fields update immediately upon save ✓ Email change allowed.	✗ Email is changed without confirmation.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
	demographic information stays current.	✓ Phone number format is validated	✗ Profile saves silently with an invalid phone format

1.2 Module 2: Medication Management

Covers creating, editing, deactivating, and viewing medications with all prescription details.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-05	[Patient] As a Patient, I want to add a new medication with its details so that my complete medication regimen is recorded.	✓ Medication is saved and appears in the medication list immediately ✓ All mandatory fields are validated ✓ Dosage form accepts Tablet, Capsule, Liquid, Injection only ✓ Frequency per day must be a positive integer	✗ Medication is saved with missing mandatory fields ✗ Frequency Per Day accepts zero or negative values ✗ End date earlier than start date is accepted
FR-06	[Patient] As a Patient, I want to edit an existing medication's details so that I can keep my prescription information accurate when dosages change.	✓ Edited medication reflects updates in all downstream schedules ✓ Confirmation dialog shown before saving changes that affect active dose schedules ✓ Audit timestamp is updated on every edit	✗ Edits are applied without confirmation when active dose schedules exist ✗ Medication name is changed to an empty string
FR-07	[Patient] As a Patient, I want to deactivate (soft-delete) a medication I no longer take so that my compliance history is preserved while the drug no longer appears in active reminders.	✓ Medication IsActive flag is set to false; it no longer triggers reminders ✓ Historical intake logs remain intact ✓ Deactivated medications are accessible in a separate 'Past Medications' view	✗ Medication is hard-deleted, removing intake logs ✗ Deactivated medication continues to generate dose reminders

1.3 Module 3: Dose Scheduling Engine

Covers auto-generation of dose schedules and browser-based notification reminders.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-08	[Patient] As a Patient, I want the system to automatically generate dose schedule entries based on the frequency I entered so that I do not have to manually create each reminder.	✓ Creating a medication with FrequencyPerDay = 3 auto-generates 3 DoseSchedule records ✓ Generated times are evenly distributed across waking hours by default ✓ Patient can override auto-generated times manually	✗ No schedules are generated after medication creation ✗ All three generated times are identical
FR-09	[Patient] As a Patient, I want to receive browser notifications for each upcoming dose so that I never miss a scheduled medication.	✓ Notification fires at the scheduled time with the drug name and dosage ✓ Notification persists until dismissed or dose is logged ✓ A missed-dose alert fires 30 minutes after the scheduled time if no intake is logged	✗ Notifications fire without the patient granting browser notification permission ✗ Missed-dose alert never fires when the patient ignores the first notification

1.4 Module 4: Intake & Compliance Logging

Covers dose event logging, compliance streak tracking, and the patient compliance dashboard.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-10	[Patient] As a Patient, I want to log each dose as Taken, Missed, or Delayed so that my compliance history accurately reflects my real medication behaviour.	✓ Intake log record is created with correct status and UTC timestamp ✓ Taken status requires ActualTakenTime to be set automatically ✓ Missed status sets ActualTakenTime to null ✓ Patient compliance streak increments on fully-compliant days and resets on any missed dose	✗ Intake log can be permanently deleted by the patient ✗ ActualTakenTime is stored for a Missed status entry ✗ A single dose can be logged twice for the same schedule slot on the same day

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-11	[Patient] As a Patient, I want to view my compliance dashboard showing today's schedule, streaks, and a ring-chart progress summary so that I can quickly assess my daily adherence.	✓ Dashboard loads in asap with all today's scheduled doses ✓ Ring chart updates in real time as doses are logged ✓ Streak counter is visually highlighted when > 7 days	✗ Dashboard shows stale data requiring a manual page refresh ✗ Compliance percentage is calculated incorrectly (e.g., delayed doses counted as missed)

1.5 Module 5: Drug Interaction Checker

Covers OpenFDA API integration, severity-rated interaction results, and saving interaction reports.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-12	[Patient] As a Patient, I want to enter a new drug name and check it against my existing medications via the OpenFDA Drug API so that I can identify dangerous interactions before filling a prescription.	✓ OpenFDA API is called with the new drug name ✓ Results show each interaction pair with a severity badge (Minor/Moderate/Major/Contraindicated) ✓ Plain-language explanation is displayed for each interaction ✓ 'Consult Your Doctor' advisory appears for Major and Contraindicated results ✓ Empty-state illustration shown when no interactions are detected	✗ Interaction check runs against an empty medication list ✗ API errors result in a blank screen with no user feedback ✗ Severity badge colours do not match the defined color system
FR-13	[Patient] As a Patient, I want to save an interaction check report so that I can reference it later or share it with my doctor.	✓ Saved report is stored with IsSaved = true and appears in Interaction Reports list ✓ Report captures QueryDrugName, ExistingDrugNames, SeverityLevel, ExplanationText, and GeneratedAt timestamp ✓ Patient can view, filter, and delete saved reports	✗ Unsaved reports are retained indefinitely without explicit save action ✗ Saved report omits the Explanation Text or timestamp

1.6 Module 6: Caregiver Portal

Covers patient-initiated caregiver linking, link approval workflow, and read-only compliance monitoring by caregivers.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-14	[Patient] As a Patient, I want to send a caregiver link request to a registered caregiver by searching their email so that I can authorize a family member to monitor my medication compliance.	✓ Patient searches caregiver by email and sends a pending link request ✓ Caregiver receives an in-app notification of the pending request ✓ Patient can cancel a pending request before it is approved ✓ Duplicate requests to the same caregiver are blocked	✗ Link is automatically approved without caregiver action ✗ Patient can send link requests to non-Caregiver role users
FR-15	[Caregiver] As a Caregiver, I want to approve or reject a patient's link request so that I control which patients I monitor.	✓ Caregiver sees a list of pending link requests with patient names ✓ Approved links set Status = Approved and record ApprovedDate ✓ Rejected links set Status = Rejected and are removed from the pending list	✗ Link request approval is processed without caregiver action ✗ Caregiver is shown the patient's medication data before approving the link
FR-16	[Caregiver] As a Caregiver, I want to view my linked patients' compliance overview, missed dose alerts, and upcoming reminders so that I can proactively support their medication adherence.	✓ Compliance overview shows 7/14/30-day adherence percentage toggle ✓ Missed dose alert log shows timestamps and medication names ✓ Upcoming reminders show the next 5 scheduled doses ✓ Data is read-only for the caregiver	✗ Caregiver can log or modify doses on behalf of the patient ✗ Caregiver can view patient data without an Approved link status

1.7 Module 7: Appointments & Refill Reminders

Covers scheduling doctor/pharmacy appointments, Zoom consultation linking, and automated refill reminder generation.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-17	[Patient] As a Patient, I want to schedule a doctor consultation, video consultation, or pharmacy visit so that I can manage all healthcare appointments in one place.	✓ Appointment is created with Type, ScheduledDateTime, Doctor/Location, and Status = Scheduled ✓ ScheduledDateTime must be a future datetime ✓ Video consultation type triggers Zoom meeting link generation ✓ Appointment appears in the unified calendar view	✗ Appointment is created with a past ScheduledDateTime ✗ Video consultation is created without a Zoom meeting link
FR-18	[Patient] As a Patient, I want refill reminders to be automatically generated based on my medication end dates so that I never run out of medication unexpectedly.	✓ RefillReminder is auto-created when a medication with an EndDate is saved ✓ Reminder fires LeadTimeDays before the RefillDueDate ✓ Patient can acknowledge the reminder, setting IsAcknowledged = true ✓ Refill reminders appear alongside appointments in the calendar view	✗ Refill reminder is not created for medications with an EndDate ✗ Reminder fires after the RefillDueDate have already passed

1.8 Module 8: Reports

Covers monthly compliance report generation, statistical accuracy, and PDF export.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-19	[Patient] As a Patient, I want to generate a monthly compliance report showing doses scheduled, taken, missed, and delayed so that I can share accurate	✓ Report is generated for the selected month/year ✓ OverallCompliancePercent is computed accurately ✓ Most-missed medication is highlighted with a pattern note	✗ Report generates with mismatched dose totals ✗ Overall Compliance Percent exceeds 100% for any patient

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
	adherence data with my healthcare provider.		
FR-20	[Patient] As a Patient, I want to export my compliance report as a PDF so that I can hand it to my physician during a consultation.	✓ PDF export button generates a professionally formatted, downloadable file ✓ PDF includes patient name, month, summary card, and medication breakdown table etc. ✓ PdfExportPath is stored in the ComplianceReport record upon successful export	✗ PDF is exported without the patient's name or report month ✗ PdfExportPath remains null after a successful export

1.9 Module 9: Doctor Management & Consultation

Covers doctor profile registration, license verification, availability management, and Admin verification workflow.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-21	[Doctor] As a Doctor, I want to register a professional profile with my license number, specialisation, qualifications, and experience so that patients can discover and book consultations with me.	✓ Profile is saved with all mandatory fields validated ✓ LicenseNumber is checked for uniqueness across all Doctor records ✓ VerificationStatus is set to Pending until Admin approves ✓ Doctor appears in patient search only after IsVerified = true	✗ Doctor profile is publicly visible before Admin verification ✗ Duplicate license numbers are accepted
FR-22	[Doctor] As a Doctor, I want to define my weekly availability slots so that patients can book appointments only during times I am available.	✓ Availability slots are saved with DayOfWeek, StartTime, and EndTime ✓ EndTime must be after StartTime - validation error shown otherwise ✓ Overlapping slots for the same day are blocked	✗ Availability slot with EndTime before StartTime is saved ✗ Overlapping slots for the same day are silently created
FR-23	[Admin] As an Admin, I want to verify or reject a doctor's	✓ Admin sees a list of doctors with	✗ Doctor can self-approve their own verification

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
	license so that only credentialed medical professionals can offer consultations on the platform.	- VerificationStatus = Pending ✓ Approving sets IsVerified = true and VerificationStatus = Verified ✓ Rejecting sets VerificationStatus = Rejected and blocks appointment booking ✓ Admin can provide a rejection reason visible to the doctor	✗ Rejected doctor can still accept new patient appointments

1.10 Module 10: Subscription & Payment Management

Covers Stripe-powered subscription billing, Free/Premium tier gating, invoice history, and payment method management.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-24	[Patient] As a Patient, I want to subscribe to the Premium tier via Stripe so that I can unlock doctor discovery, appointment booking, and video consultations.	✓ Stripe checkout session is created and redirects to the payment page ✓ Successful payment sets Subscription Status = Active and Tier = Premium ✓ StripeSubscriptionId is stored in the Subscription record ✓ Only one Active subscription exists per user at any time	✗ Premium features are accessible before payment is confirmed ✗ A user can have two Active subscriptions simultaneously
FR-25	[Patient] As a Patient, I want to view my invoice history and manage my payment method so that I have full visibility over my billing.	✓ Invoice list shows Amount, Status, DueDate, and PaidDate for all past invoices ✓ Patient can update their Stripe payment method ✓ Overdue invoices are visually highlighted	✗ PaidDate is shown as non-null for an invoice with Status ≠ Paid ✗ Invoice list is empty immediately after a successful payment

1.11 Module 11: Admin Panel

Covers admin-level user management, doctor verification, and consultation oversight.

ID	User Story	Acceptance Criteria ✓	Rejection Criteria ✗
FR-26	[Admin] As an Admin, I want to view, filter, and manage all user accounts so that I can maintain platform integrity and respond to complaints.	✓ Admin can search users by name, email, and role ✓ Admin can suspend a user, blocking their login immediately ✓ Admin can permanently delete a user after confirmation (blocked if active subscriptions exist)	✗ Admin can delete a user with an active subscription without a warning ✗ Admin search returns no results for valid email queries
FR-27	[Admin] As an Admin, I want to oversee and cancel any consultation so that I can intervene in cases of flagged or inappropriate sessions.	✓ Admin can view all consultations with filters for Status and IsFlagged ✓ Admin can cancel a consultation, setting Status = Cancelled ✓ Cancellation is blocked and replaced with a restriction notice if a ConsultationNote exists	✗ Admin cancels a completed consultation that has clinical notes ✗ Flagged consultations are not visually distinguishable from unflagged ones

Section 2:

Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints the MediAlert system must satisfy. They are expressed in a user-story-oriented format with measurable acceptance criteria, a target metric, and clear rejection conditions.

ID	User Story / Requirement	Acceptance Criteria ✓	Metric / Target	Rejection Criteria X
NFR-01	Performance As a Patient, I want the dashboard to load asap so that I can check my doses quickly, even on a mobile connection.	✓ Dashboard renders all today's doses within a minimum time asap on a standard broadband connection ✓ Ring-chart updates reflect new intake log in < 500 ms	≤ 2 s (95th percentile)	- X Dashboard takes > 5 seconds to load on a standard connection - X Ring-chart requires a manual refresh to update
NFR-02	Performance As a Patient, I want the OpenFDA drug interaction check to return results asap so that the checker feels responsive and useful.	✓ API call and result rendering complete asap for a typical 5-drug regimen ✓ Loading spinner is shown during API call	≤ 5 s (90th percentile)	- X Results take > 10 seconds with no loading indicator - X Timeout occurs with no error message shown
NFR-03	Security As a Patient, I want all my health data transmitted with JWT authentication so that my sensitive medical information is protected from interception.	✓ JWT tokens expire after 24 hours of inactivity ✓ Passwords are stored as hashes	TLS 1.2+, JWT TTL ≤ 24 h	X Passwords are logged or stored in plaintext
NFR-04	Security As a Patient, I want role-based access control	✓ Doctor endpoints reject patients without an	Zero unauthorised data access incidents	X Caregiver can read patient medication data without

ID	User Story / Requirement	Acceptance Criteria ✓	Metric / Target	Rejection Criteria X
	enforced at the API layer so that a Caregiver cannot access medication data they are not approved to view.	Approved PatientDoctorLink ✓ Admin endpoints return error for non-admin JWT tokens		an Approved link X A Patient JWT is accepted by an Admin-only endpoint
NFR-05	Reliability As a Patient, I want the reminder notification system to deliver dose alerts reliably so that I do not miss critical medication times due to system downtime.	✓ Scheduled notifications are queued and delivered within 60 s of trigger time ✓ Missed-dose alerts are generated even if the patient's browser was closed when the dose was due	99.5% notification delivery within 60s	X Notifications are silently dropped if the browser is inactive X Reminder system fails without any error logged
NFR-06	Usability As an elderly Patient, I want the UI to use high-contrast colours and accessible font sizes so that I can read and interact with MediAlert without strain.	✓ All text meets WCAG 2.1 AA contrast ratio ($\geq 4.5:1$) ✓ Default body font size is $\geq 14\text{pt}$ ✓ All interactive elements have a minimum touch target of $44 \times 44\text{px}$	WCAG 2.1 AA compliance	X Any screen fails the WCAG AA contrast check X Touch targets on mobile are $< 44\text{px}$
NFR-07	Usability As a Patient, I want the application to be fully responsive on mobile devices so that I can manage my medications from my smartphone.	✓ All five primary screens render correctly on 375 px (iPhone SE) and 768 px (iPad) viewports ✓ No horizontal scrollbars appear on mobile viewports ✓ All buttons and inputs are usable without zooming	0 layout breaks at 375 px+	X Any screen produces a horizontal scrollbar at 375 px X Text or buttons overflow their containers on mobile
NFR-08	Scalability As a System, the database	✓ All critical queries (compliance)	< 200 ms for indexed queries	X Query time increases

ID	User Story / Requirement	Acceptance Criteria ✓	Metric / Target	Rejection Criteria X
	must handle a growing user base efficiently so that response times do not degrade as more patients are added.	- dashboard, medication list) execute asap ✓ Foreign key indexes are present on all FK columns		- proportionally with table size (full table scan) - X FK columns lack database indexes
NFR-09	Data Integrity As a Patient, I want the system to enforce referential integrity so that deleting a medication never leaves orphaned dose schedules or intake logs.	✓ Deleting a Medication cascade to DoseSchedule and RefillReminder ✓ Deleting a DoseSchedule cascades to IntakeLog ✓ Deleting a Consultation is blocked when consultation note records exist	Zero orphaned records after any delete	- X Orphaned DoseSchedule records remain after a Medication is deleted - X Orphaned IntakeLogs are not linked to any DoseSchedule
NFR-10	Maintainability As a Developer, I want the application to be organised into clearly separated modules (Auth, Medication, Reports, etc.) so that each module can be updated independently without breaking others.	✓ Each bounded context (Identity, Clinical, Financial, Compliance) is isolated in its own code module ✓ No circular dependencies exist between modules ✓ Database migrations are version-controlled	0 circular module dependencies	- X A change in the Medication module requires edits to the Authentication module - X Database schema changes are applied manually without migration scripts
NFR-11	Privacy As a Patient, I want my health data stored securely and not shared without my consent so that my medical information remains private.	✓ HealthProfile data is accessible only by the owning Patient and explicitly approved Caregivers/Doctors ✓ Admin has no direct FK access to clinical health records ✓ All third-party API calls (OpenFDA,	Zero unapproved cross-role data access	- X HealthProfile is returned in a general user query with no role check - X Full medication lists are transmitted to third-party APIs unnecessarily

ID	User Story / Requirement	Acceptance Criteria ✓	Metric / Target	Rejection Criteria X
		Stripe, Zoom) transmit only the minimum necessary data		

***** The End *****

